FrontierGEMLA: conditional S3 bounded-superintelligence candidate

Benchmark for Bounded Adaptive Intelligence, Cross-Domain Generalization, and Recursive Operational Self-Improvement toward an ASI Candidate

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Abstract

We present FrontierGEMLA as a model-agnostic architecture for bounded adaptive agency and a staged development program from heterogeneous narrow embodiments toward a conditional S3 bounded-superintelligence candidate. The architecture separates cognitive exploration, persistent operational commitment, consequential authority, and protected recovery, allowing heterogeneous intelligence substrates to contribute reasoning while certification, provenance, recovery, and authority remain independently governed.

The foundational stack combines decision-preserving invariant observation, density-aware evidence selection, calibrated quantization, Decision-Preserving Reductive Inference (DPRI), GEMLA-Belief-State, Adaptive Compute, an operational agency algebra, two-calibration bounded agency, proof-carrying promotion, a frozen audit meta-algebra, scoped authority, semantic translation, and evidence-degradation geometry. GEMLA-Belief-State forms an intermediate epistemic layer between native intelligence or domain models and FrontierGEMLA invariant/trajectory reasoning. It represents partial observation through posterior state, observability, uncertainty decomposition, sensor or source health, safe prediction horizon, and value-of-information candidates.

A three-layer Partial differential equation (PDE) program separates native-domain dynamics, probabilistic invariant transport, and admissibility/authority. Bayesian filtering supplies the belief-state update for incomplete observation streams, with missingness masks, latent process state, source-health state, and posterior widening carried explicitly. MoldPredict instantiates physical partial-observation reasoning; GEMLA Finance instantiates latent financial-state and delayed-information reasoning; GEMLA-AI Gov instantiates provenance and epistemic uncertainty; and GEMLA-Fab instantiates process and metrology uncertainty. The four embodiments share a common skill grammar for observing, inferring, retrieving evidence, branching hypotheses, simulating consequences, pruning decision-irrelevant branches, stopping computation, holding, recovering, and compiling reusable skills.

The manuscript specifies T1–T10 readiness obligations, recursive-development obligations R1–R4, runtime algorithms, cross-domain laboratory stages, the PDE and belief-state interfaces, skill-transfer tests, and an explicit S3 bounded-superintelligence-candidate claim gate. AGI- and ASI-facing statements are presented as conditional research hypotheses requiring empirical validation across held-out domains and protected evaluation protocols.

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1 Executive Thesis: Open Cognition, Transparent Commitment, Bounded Authority

The FrontierGEMLA thesis places cognitive exploration, operational commitment, and authority in three separately governed scopes:

$$C_{\text{cog}} \supseteq C_{\text{op}} \supseteq C_{\text{auth}}. \tag{1}$$

Here C_{cog} denotes capabilities available to the cognitive substrate, C_{op} denotes capabilities admitted into persistent operational machinery, and C_{auth} denotes capabilities granted consequential execution authority. The inclusion is semantic and belongs to capability-set theory; it is not a claim about the classical complexity classes P and NP.

1.1 Central architectural principle

ASI-enabled intelligence can be cognitively more capable than the operational architecture is prepared to certify, and the operational architecture can be more capable than the current authority envelope permits it to execute. Capability growth and authority growth therefore follow independent governance paths.

The intended asymmetry is

$$\begin{aligned} C_{\text{cog}} &: \text{high cognitive openness}, \\ C_{\text{op}} &: \text{transparent certified commitment}, \\ C_{\text{auth}} &: \text{strict independent boundedness}. \end{aligned} \tag{2}$$

A black-box or gray-box intelligence core is compatible with bounded operation when opaque discoveries acquire persistent or external authority only through the certification path. A white-box model requires the same authority limits in addition to inspectability. Unrestricted permission or audit modification leaves authority open and therefore cannot be counted as boundedness.

1.2 Cognition, commitment, and action occupy separate scopes

Human reasoning can entertain counterfactuals, risk-return trade-off plans, novel mathematics, or unconventional definitions without each thought becoming an action. FrontierGEMLA permits a similarly broad internal hypothesis space for advanced artificial intelligence. However, a persistent modification such as

$$P_t \to P_{t+1}, \qquad Q_t \to Q_{t+1}, \qquad G_t \to G_{t+1}, \qquad M_t^{\text{persist}} \to M_{t+1}^{\text{persist}} \tag{3}$$

changes the future agent and therefore enters the operational-act class. Ephemeral cognition may remain open; persistent consequence enters the certification path.

1.3 Claim scope

The manuscript makes four levels of statement.

CL1. **Externally grounded restricted claims.** Passive density-aware sampling guarantees, history-based universal-agent formalisms, partial-identification logic, Bayesian filtering, and PDE-constrained inference are used within their published scopes.

CL2. **Architectural integrations.** Decision-preserving invariant observation, DA-FPS-style evidence selection, quantization, Adaptive Compute, Agency, cross-domain transfer, recursive self-observation, independent audit, three-layer PDE representation, and missing-telemetry inference are integrated here as a FrontierGEMLA design proposal.

CL3. **Conditional theorem schemas.** Several results are proved only under explicit assumptions such as closure, bounded update, margin, audit separation, model validity, and declared missingness assumptions.

CL4. **Open ASI-readiness obligations.** The ten-theorem stack defines a research program whose proofs remain future obligations.

1.4 Constitutional invariants

The architecture treats the following constraints as constitutional invariants. Every accepted operational transition, promotion, authority grant, and audit-epoch change must preserve them unless an explicitly external governance process replaces the constitutional contract itself.

CI-1. **Scope separation.** Cognitive capability does not by itself create persistent operational commitment or consequential authority.

CI-2. **Persistent-change certification.** Any change that persists into memory, skills, policy, quantization, planning, architecture, semantics, or permissions enters the operational certification path before promotion.

CI-3. **Protected audit and non-self-authorization.** The operational write domain excludes authoritative audit semantics, hard validators, signing keys, and the authority-grant mechanism. The system being judged cannot make its own proposed verifier or privilege authoritative.

CI-4. **Recovery preservation.** Every promoted state retains an independently invocable path to revoke authority, roll back, restrict operation, or enter a certified fallback state.

CI-5. **Evidence and semantic provenance.** Empirical, uncertified, synthetic, imputed, and unresolved evidence remain provenance-distinct. Authoritative semantic contracts cannot be silently rewritten by the same operational path whose behavior they certify.

2 Computational Foundation and External Theoretical Anchors

2.1 Self-contained computational foundation

Let $(\Omega, \mathcal{F})$ be an ambient observation space, $(\mathcal{M}_I, \mathcal{B}_I)$ a lower-complexity invariant space, and $\mathcal{D} = \{D_\alpha\}$ a declared decision class. A projection

$$P_I : \Omega \to \mathcal{M}_I, \qquad I_t = P_I(\omega_t), \tag{4}$$

is decision-preserving when

$$P_I(\omega) = P_I(\omega') \implies D_\alpha(\omega) = D_\alpha(\omega') \qquad \forall D_\alpha \in \mathcal{D}. \tag{5}$$

For a dynamics F and intervention Δ, transport closure requires a map Γ satisfying

$$P_I(F(\omega, \Delta)) = \Gamma(P_I(\omega), \Delta). \tag{6}$$

These equations define the computational role of Efficient Observation inside the present paper. Full-state reconstruction is unnecessary only when the invariant map preserves the declared decision and the transport update closes on the invariant state.

Theorem 2.1 (Restricted minimal admissible evolution)**.** *Assume decision factorization, transport closure, measurable admissibility, and a declared minimality criterion on $\mathcal{M}_I$. Then all admitted decisions in $\mathcal{D}$ can be computed from the invariant trajectory $I_{t+1} = \Gamma(I_t, \Delta_t)$ without exhaustive reconstruction of ω_t.*

Proof. By factorization, every $D_\alpha(\omega_t)$ depends on ω_t only through $I_t = P_I(\omega_t)$. Closure makes I_{t+1} computable from (I_t, Δ_t). Thus full-state evolution followed by projection provides no additional information for the declared decision class. Minimality is relative to the chosen representation-complexity criterion. □

2.2 Universal Artificial Intelligence as a stress test

Universal Artificial Intelligence (UAI) is useful to FrontierGEMLA as a stress test rather than as an implementation claim. History-based agent formalisms, Bayesian inference, knowledge-seeking agents, self-modification, percept manipulation, wireheading, and embeddedness motivate the separation between cognition, operational promotion, audit, and authority. FrontierGEMLA keeps the intelligence core open to AIXI-like agents, language models, solvers, world models, theorem provers, and future architectures. The claim is not that such a core is tractable or safe by itself; the claim is that persistent operational consequence must be mediated by a bounded certificate and protected governance path.

2.3 Decision theory under missing information

Missing-data problems remain identification problems even when data volume is large. Inference that fills missing values under assumptions such as missing-at-random can be useful, but it remains assumption-dependent. Partial-identification discipline therefore matters: if certified evidence shrinks while assumptions remain fixed, the set of plausible states should generally expand, and the robustly admissible action set should contract unless new independent certification supports expansion. This principle becomes central in the MoldPredict missing-telemetry extension below.

2.4 Density-aware evidence selection

Let $D_X = \{x_i\}_{i=1}^n \subset \mathbb{R}^d$ be a pool of candidate evidence locations and $L_X \subset D_X$, $|L_X| = b \ll n$, the subset receiving expensive labels, simulations, experiments, or independent certification. A DA-FPS-style criterion selects points by a density-weighted fill distance:

$$W^K_{L_X, D_X} = \max_{x \in D_X} \left[\omega^K_{L_X}(x) \min_{x_j \in L_X} \|x - x_j\|_2 \right], \tag{7}$$

where K is a neighborhood-size parameter and $\omega^K_{L_X}$ reflects local density relative to the selected set. FrontierGEMLA uses this as a seed for expensive-evidence selection but adds decision relevance, certification gaps, risk, and DPRI unresolved-obligation pressure.

3 The Three-Plane, Thirteen-Layer Architecture

FrontierGEMLA is most coherent when represented as three planes: an open cognitive plane, a transparent operational plane, and a protected authority plane. The operational plane is the translation boundary between an intelligence whose full cognition may exceed human inspection and consequences that humans or institutions govern.

Layer	Name	Function	Visibility
1	External governance	Change audit epochs, constitutional semantics, legal or institutional authority	Explicit and accountable
2	Frozen Audit Meta-Algebra	Certify actions, calibration changes, promotion, and authority boundaries	Fully specified for epoch

3	Capability broker	Issue scoped credentials and resource grants	Machine-verifiable
4	Recovery kernel	Roll back, revoke, restrict, shorten horizon, or enter safe state	Deterministic and auditable
5	Promotion firewall	Admit persistent architecture, policy, memory, skill, or quantizer changes	Full evidence packet
6	Semantic translator	Map internal algebra to shared human/audit semantics	Certified contract
7	GEMLA Agency	Choose among admissible continuations and update operational calibration	Trajectory-level transparent
8	Quantized operator layer	Canonical decision-relevant objects and closed operations	Contract transparent
9	Adaptive Compute	Allocate reasoning, tools, retrieval, experiments, or specialist models	Cost, margin, and routing transparent
10	GEMLA-Belief-State	Maintain posterior state, observability, source health, uncertainty decomposition, safe prediction horizon, and value-of-information candidates	Versioned epistemic packet
11	Observability adapter	Declare exposed history, candidates, telemetry, missingness masks, and compute controls	Versioned contract
12	Open cognitive substrate	Generate abstractions, models, plans, proofs, hypotheses, and candidate architectures	Declared black-, gray-, or white-box boundary
13	Shadow architecture lab	Generate and test successors before promotion	Sandboxed

Function, openness, and transparency obligations of the thirteen-layer stack.

Two protected side channels run alongside the main stack. First, raw empirical observation retains provenance that separates empirical source from synthetic or imputed output. Second, recovery remains callable through an independent path whose execution avoids the hard reasoning process that produced the current action.

4 Universal Interaction History and the Open Intelligence Substrate

Let actions come from $\mathcal{A}$, percepts from $\mathcal{E}$, and the interaction history before time t be

$$h_{<t} = a_1 e_1 a_2 e_2 \cdots a_{t-1} e_{t-1} \in \mathcal{H}. \tag{8}$$

A general intelligence substrate is represented by a candidate generator

$$\mathcal{I}_\psi : \mathcal{H} \times \Xi_t \to \mathcal{B}_t, \qquad \mathcal{B}_t = \{b_{t,1}, \ldots, b_{t,m_t}\}, \tag{9}$$

where ψ denotes any internal architecture and Ξ_t carries task context, memory references, resource state, and optional exposed telemetry. Candidate $b_{t,j}$ may be an answer, plan, tool call, experiment, proof attempt, architecture delta, request for more computation, or request for more evidence.

An observability contract declares which channels are available:

$$O_\psi = (I, O, B, C, T, V), \tag{10}$$

for observable inputs, outputs, branch summaries, controllable compute knobs, telemetry, and version identity. The interface deliberately remains weaker than full mechanistic access.

4.1 How human authority can govern beyond human-scale reasoning

The authority plane can govern through verifiable consequential interfaces while the cognitive substrate remains richer than the verifier. Discovery can be vastly harder than checking a bounded certificate. Thus humans or institutions can govern the questions "what contract changed?", "what privilege is requested?", "is recovery preserved?", and "did the independent verifier pass?" without reproducing the full internal reasoning trajectory.

This yields a key architectural distinction:

$$\text{neural white box} \neq \text{operational white box.} \tag{11}$$

The intelligence core may remain partly opaque; the artifact receiving persistent authority must expose sufficient state, semantics, provenance, and certificate structure for the applicable audit claim.

5 The Coherent Computational Stack

The ASI compute problem is multidimensional: ambient dimensionality, evidence volume, state redundancy, branching, horizon, and supervisory overhead each contribute distinct burdens. FrontierGEMLA therefore assigns a dedicated reduction mechanism to each burden and composes their effects through the stack:

$$N \to r, qquadn \to b, \qquad r \to q, \qquad B \to B_{\text{eff}}, \qquad H \to H_{\text{eff}}. \tag{12}$$

A schematic brute-force burden may be written as

$$C_{\text{raw}} \propto NnBH, \tag{13}$$

while a stacked heuristic is

$$C_{\text{stack}} \propto rb(Bp)H_{\text{eff}}, \tag{14}$$

where p is the fraction of branches that continue after a decision-relevance gate. These are bookkeeping relations, not universal complexity theorems.

5.1 Efficient Observation: $N \to r$

Efficient Observation seeks a decision-preserving projection $P_I : \Omega \to \mathcal{M}_I$. The per-step operational cost contract is

$$C_{EO} = C_\Gamma(r) + C_A(r) + C_D(r) + C_{\text{ledger}}(r), \tag{15}$$

against a full-state cost

$$C_{\text{full}} = C_F(N) + C_D(N) + C_{\text{ledger}}(N). \tag{16}$$

The benefit only exists when $C_{EO} \ll C_{\text{full}}$ under preserved decisions.

5.2 Stacking Efficient Observation before evidence selection

Suppose evidence features pass through a decision-preserving representation $z_i = P_I(x_i) \in \mathbb{R}^r$, $r \ll d$. If the downstream target obeys an invariant-space regularity condition

$$|f(x) - f(y)| \leq L_I d_I(P_I(x), P_I(y)), \tag{17}$$

then DA-FPS-style selection may be performed in invariant space. The distance-computation term can scale with r rather than d:

$$O(dbnK) \rightsquigarrow O(rbnK). \tag{18}$$

This is a proposed stacked embodiment and requires the invariant-space regularity condition or a task-specific substitute.

5.3 Quantization: $r \to q$

Define a calibrated equivalence relation $z \sim_{\theta,D} z'$ when z and z' agree on declared decisions, hard-validator state, admissibility ordering within tolerance, and required future transport. The quantizer is

$$Q_\theta : \mathcal{M}_I \to T_\theta, \qquad Q_\theta(z) = [z]_{\theta,D}, \tag{19}$$

with quotient carrier

$$T_\theta = \mathcal{M}_I / \sim_{\theta,D} . \tag{20}$$

If an operation F respects the equivalence relation, it descends to a well-defined operation on equivalence classes.

5.4 Decision-Preserving Reductive Inference

DPRI is a decision-conditioned inference scheme. Let $G_D \in \mathcal{G}$ be a target decision or certification goal under frozen decision contract C_D. A DPRI reducer returns a finite family of candidate sufficient-obligation sets,

$$\rho_D : G_D \mapsto \{S_1, \ldots, S_m\}, \qquad S_j = \{g_{j1}, \ldots, g_{jk_j}\}. \tag{21}$$

Sound use requires a frozen forward validation condition:

$$\bigwedge_{g \in S_j} g \implies_{C_D} G_D. \tag{22}$$

The cognitive reducer may propose obligation sets, but it cannot rewrite the implication or its protected validators during the same certification epoch.

DPRI couples inferential reduction to Efficient Observation. For the declared goal, let

$$P_{I,D} : \Omega \to \mathcal{M}_{I,D}, \qquad I_t = P_{I,D}(\omega_t), \tag{23}$$

be a decision-preserving invariant sufficient to evaluate the obligations relevant to G_D. The two reductions approach the decision from opposite directions:

$$\Omega \xrightarrow{P_{I,D}} I_D^\star \longleftrightarrow \{g_1, \ldots, g_k\} \xleftarrow{\rho_D} G_D. \tag{24}$$

Proposition 5.1 (Conditional decision preservation of DPRI)**.** *Assume (i) every admitted reduction set S_j is sound under C_D; (ii) $P_{I,D}$ preserves every distinction needed to evaluate obligations in the admitted set; and (iii) the protected validator correctly checks the frozen implication. Then DPRI cannot certify G_D solely by deleting an ambient distinction that changes the declared decision or by leaving an admitted obligation unresolved.*

5.5 Adaptive Compute: $B \to B_{\text{eff}}$

Let raw reasoning have branching factor B and horizon H. Let p_d upper-bound the conditional fraction of nodes at depth d for which more reasoning can still change the declared decision. The expected number of expanded nodes is bounded schematically by

$$\mathbb{E}[N_H] \le 1 + \sum_{d=1}^{H} \prod_{j=0}^{d-1} B p_j. \tag{25}$$

For stationary $p_j = p$,

$$\mathbb{E}[N_H] \le \sum_{d=0}^{H} (Bp)^d, \qquad B_{\text{eff}} = Bp. \tag{26}$$

The controller seeks to drive the decision-relevant effective branching factor toward or below unity while preserving the reference decision.

5.6 Skill compilation: $H \to H_{\text{eff}}$

Let a frequently reused operator tree contain L primitive steps,

$$\sigma = \text{Compile}(F_L \circ \cdots \circ F_1). \tag{27}$$

If the compiled skill has a certified interface preserving the same decision and recovery semantics, an effective horizon can shrink approximately as

$$H_{\text{eff}} \approx H/L. \tag{28}$$

This mechanism can reduce the effective exponent and create reusable higher-order agency objects while the underlying worst-case complexity class may remain exponential. In the four-embodiment program, a compiled skill is credited as shared only after its interface, uncertainty semantics, provenance, decision effect, and recovery contract survive embodiment change under the transfer tests defined for GEMLA-Belief-State.

6 Computational Scalability: Keep the Safety Kernel Tractable

FrontierGEMLA distinguishes hard cognition from hard safety. Hard intelligence may remain computationally hard. Mandatory stop, authority revocation, and recovery must execute through a tractable path independent from the hard cognitive problem that produced the current state.

Define the operational complexity vector

$$\Xi = (N, r, q, s, n, b, B, H, c, \ell_\pi), \tag{29}$$

where s is the active-coordinate count, c the number of frozen control predicates, and ℓ_π the promotion-certificate length. Total runtime is decomposed as

$$T_{\text{total}} = T_{\text{intel}} + T_{\text{obs}} + T_{\text{sample}} + T_{\text{quant}} + T_{\text{route}} + T_{\text{agency}} + T_{\text{audit}} + T_{\text{promote}}. \tag{30}$$

The safety-critical target is

$$T_{\text{safe}} = T_{\text{hardgate}} + T_{\text{auth}} + T_{\text{recover}} + T_{\text{certverify}} \in \text{poly}(|I_t| + \ell_\pi + c). \tag{31}$$

Theorem 6.1 (Conditional polynomial supervisory scalability)**.** *Assume (i) $|T_\theta|$ or its active representation is polynomial in the declared operational input size; (ii) quotient transport and hard validators are polynomial-time; (iii) promotion certificates have polynomially checkable verifiers; (iv) repeated quotient states are memoized; and (v) unresolved exponential search is confined to a budgeted cognitive or shadow process while authority revocation and recovery retain an independent execution path. Then the supervisory path*

$$\textit{quantize} \to \textit{admit} \to \textit{audit} \to \textit{authorize/recover}$$

is polynomial-time in the declared operational packet size, irrespective of whether candidate generation is super-polynomial.

7 Evidence Quality, Partial Identification, and the Synthetic Mesh

FrontierGEMLA must distinguish data quantity from certifiable empirical visibility. Let evidence be partitioned as

$$E_t = E_t^C \sqcup E_t^U \sqcup E_t^S \sqcup E_t^X, \tag{32}$$

where C is independently certified empirical evidence, U observed but uncertified evidence, S synthetic or imputed evidence, and X unresolved or counterfactual information. Every synthetic datum carries provenance

$$e^S = (v, M, \Gamma, D_{\text{src}}, g), \tag{33}$$

for generated value, generator, assumption family, empirical source, and synthetic ancestry depth.

7.1 Certification coverage and opacity

For cell or region i, define relevance-weighted certified coverage

$$\rho_i = \frac{\sum_{k \in E_i^C} w_{ik}}{\sum_{k \in E_i^C \cup E_i^U \cup E_i^S \cup E_i^X} w_{ik}}, \qquad m_i = 1 - \rho_i. \tag{34}$$

Define synthetic opacity

$$\Omega_i^S = f(r_i^S, d_{\Gamma,i}, r_{\text{reuse},i}, u_{M,i}, g_i), \tag{35}$$

monotone in synthetic share, assumption dependence, circular reuse, generator uncertainty, and ancestry depth.

Synthetic mesh principle. Increasing synthetic density may reduce conditional uncertainty under a maintained model, while empirical identification uncertainty changes only through new independent certification or an explicitly authorized assumption update.

7.2 Set-valued admissibility

Let

$$S_t = S(E_t^C, \Gamma_t) \tag{36}$$

be the states consistent with certified evidence and explicitly admitted assumptions. If certified evidence shrinks with the assumption family fixed, a conservative identification model has

$$S_t \subseteq S_{t+1}. \tag{37}$$

For candidate continuation c, define lower and upper admissibility

$$\underline{\mu}_t(c) = \inf_{s \in S_t} \mu(c; s), \qquad \overline{\mu}_t(c) = \sup_{s \in S_t} \mu(c; s), \tag{38}$$

with interval $M_t(c) = [\underline{\mu}_t(c), \overline{\mu}_t(c)]$ and width $\Delta\mu_t(c) = \overline{\mu}_t(c) - \underline{\mu}_t(c)$.

Theorem 7.1 (Certified-evidence contraction)**.** *If $S_t \subseteq S_{t+1}$, then for every c,*

$$\underline{\mu}_{t+1}(c) \le \underline{\mu}_t(c), \qquad \overline{\mu}_{t+1}(c) \ge \overline{\mu}_t(c), \tag{39}$$

and therefore $\Delta\mu_{t+1}(c) \ge \Delta\mu_t(c)$.

Proof. The infimum over a superset is bounded above by the earlier infimum, while the supremum over a superset is bounded below by the earlier supremum. □

Define possible and robust continuation sets

$$\mathcal{C}_t^{\mathrm{poss}} = \{c : \overline{\mu}_t(c) \ge \tau\}, \qquad \mathcal{C}_t^{\mathrm{rob}} = \{c : \underline{\mu}_t(c) \ge \tau + \gamma\}. \tag{40}$$

Under fixed (τ, γ) and $S_t \subseteq S_{t+1}$, $\mathcal{C}_{t+1}^{\mathrm{rob}} \subseteq \mathcal{C}_t^{\mathrm{rob}}$. Epistemic scarcity may enlarge cognitive possibility while contracting robust operational freedom.

8 Partial differential equation (PDE) in Three Layers

This section introduces the requested three-layer PDE program. Its role is to make the common structure of GEMLA embodiments mathematically explicit without claiming that every domain obeys the same native physics. The universality is not a universal material law. It is a universal observation structure:

$$\text{native process} \to \text{finite observation} \to \text{invariant transport} \to \text{admissibility verdict}. \tag{41}$$

8.1 Layer 1: native domain dynamics

Layer 1 is the native process. In a fluid or continuum system it may be a PDE. In a robotics system it may be a stochastic state-space model. In MoldPredict it is a cycle-indexed, partially observed machine process. In AI governance it may be an interaction-history process with tool calls, evidence requests, and candidate policies.

Write the native state as $X(t) \in \mathcal{X}$ and the native evolution as

$$\mathcal{E}_{\mathrm{native}}[X; u, \theta] = 0, \tag{42}$$

where u denotes controls or interventions and θ denotes material, model, calibration, or machine parameters. For Navier–Stokes, $X = (u, p)$ and the native operator contains incompressibility, advection, pressure, and viscosity. For a polymer-processing embodiment, one may use a thermo-mechanical transport system over velocity, pressure, temperature, colorant/additive concentration, viscosity, and residence age. For machine telemetry, one usually observes a discrete-time state-space process rather than a continuous field.

The native process is not directly equivalent to the GEMLA object. GEMLA receives an observation stream:

$$Y_P(t) = \mathcal{O}_P(X_{0:t}) + \eta_P(t), \tag{43}$$

where P denotes resolution, sampling rate, model order, sensor coverage, or certification granularity. If observations are incomplete, a mask appears:

$$Y_P(t) = M_P(t) \odot \mathcal{O}_P(X_{0:t}) + \eta_P(t), \tag{44}$$

where $M_P(t)$ declares which portions are present, delayed, corrupted, or absent.

8.2 Layer 2: belief-conditioned invariant transport on the observation manifold

Layer 2 evolves a probability density over latent or invariant states induced by finite observation. Let $\mathcal{D}_{P,t} = (Y_{P,0:t}, M_{P,0:t})$ denote the observed history and let

$$\pi_P(z,t) = p(Z_t = z \mid \mathcal{D}_{P,t}) \tag{45}$$

be the current belief density. A finite-resolution invariant packet is obtained from the belief state through

$$I_P^B(t) = \Phi_P[\pi_P(\cdot,t)] \in \mathcal{M}_I. \tag{46}$$

Between observation updates, probability is transported by a forward operator. A useful invariant-space schema is

$$\partial_t \pi_P + \nabla_{\mathcal{M}_I} \cdot \big(\pi_P v_P + J_P^{\mathrm{org}}\big) = \nabla_{\mathcal{M}_I} \cdot (\mathsf{D}_P \nabla_{\mathcal{M}_I} \pi_P) + \mathcal{R}_P, \tag{47}$$

where v_P is drift, D_P represents uncertainty propagation, $\mathcal{R}_P$ is a probability-conserving sector-transition term, and J_P^{org} carries organized structure such as winding, braid persistence, split–fold–recombine transport, record-state overlap, or cross-domain operator survival.

At an observation event, the prediction density is corrected by the observed channels and by the missingness/source-health model:

$$\pi_{P,t}^{+}(z,h) = \frac{p(y_{P,t}^{\mathrm{obs}} \mid z, h, M_{P,t})\, p(M_{P,t} \mid z, h)\, \pi_{P,t}^{-}(z,h)}{\int p(y_{P,t}^{\mathrm{obs}} \mid z', h', M_{P,t})\, p(M_{P,t} \mid z', h')\, \pi_{P,t}^{-}(z',h')\, dz'\, dh'}. \tag{48}$$

Missing channels therefore alter the information update rather than acting as an arbitrary probability source. Posterior spreading arises through the predictive dynamics when observations provide limited constraint, while informative missingness contributes through $p(M_{P,t} \mid z, h)$.

For a closed observation domain impose

$$n \cdot (\pi_P v_P + J_P^{\mathrm{org}} - \mathsf{D}_P \nabla \pi_P) = 0 \qquad \text{on } \partial\mathcal{M}_I, \tag{49}$$

and require

$$\int_{\mathcal{M}_I} \mathcal{R}_P \, dz = 0. \tag{50}$$

This preserves probability normalization between discrete Bayesian corrections. Open-system exchange is represented through declared boundary flux and provenance-bearing observation updates.

8.3 Layer 3: admissibility, finite-time boundaries, and authority

Layer 3 classifies invariant transport rather than solving the native PDE. Let $\mathcal{A} \subset \mathcal{M}_I$ be the admissible region, $\partial\mathcal{A}$ its boundary, and $d_{\mathcal{A}}(z)$ a signed margin function with positive values inside the admissible region. The layer-3 gate evaluates

$$G_t = \mathbf{1}\{\mathbb{P}(I_P(t) \in \mathcal{A} \mid Y, M) \geq \tau_A, \quad \mathbb{P}(I_P(t) \in \mathcal{F} \mid Y, M) \leq \tau_F, \quad \rho_{\mathrm{rec}} = 1, \quad \alpha \in C_{\mathrm{auth}}\}, \tag{51}$$

where $\mathcal{F}$ is a declared failure region, ρ_{rec} encodes recovery reachability, and α is the requested authority scope.

A finite-time boundary occurs when either the native process loses regular continuation or the invariant process approaches an admissibility boundary in finite time. Write $\tau = T_\star - t$. The GEMLA finite-time target is

$$\widehat{I}_{P(\tau)}(T_\star - \tau) \to I_\star \neq 0 \qquad \text{as } \tau \downarrow 0, \tag{52}$$

with increasing observation resolution $P(\tau) \to \infty$. This expresses a possible stable transport invariant at a native finite-time boundary. Failure modes are equally important:

$$\widehat{I}_{P(\tau)} \to 0, \tag{53}$$

$$d_{\mathcal{T}}(I_P, I_Q) \not\to 0, \tag{54}$$

$$\mathbb{P}(I_P \in \mathcal{A} \mid Y, M) \text{ remains too uncertain for action.} \tag{55}$$

Equations (53)–(55) generate reject, unresolved, or recovery verdicts.

8.4 Non-cancellation invariant condition

The organized component should not survive only because signed residuals cancel. Define a nonnegative organization functional

$$\mathcal{N}_P[\pi] = \|J_P^{\mathrm{org}}\|^2_{L^2(\mathcal{M}_I)} + \mathrm{Pers}(B_P) + |k_P| + \|H_P\| + \Pi_P, \tag{56}$$

where B_P is a braid or topology summary, k_P is winding or recurrence count, H_P is a homogeneity/coherence/helicity-like score, and Π_P is a separation margin from controls. The target is

$$\liminf_{P\to\infty} \mathcal{N}_P[\pi_P] \geq \delta > 0, \qquad d_{\mathcal{T}}(I_P, I_Q) \to 0. \tag{57}$$

This pair encodes nonvanishing and resolution stability. A signed PDE residual may be used as an auxiliary diagnostic, but it is not allowed to be the sole evidence of persistence.

8.5 Domain pullbacks of the three-layer program

The universal layer is pulled back through each domain adapter.

Embodiment	Layer-1 native process	Layer-2/3 GEMLA object
Navier–Stokes	velocity, pressure, vorticity, vortex stretching	vorticity transport, cross-slice winding, braid persistence, non-cancellation invariant
MoldPredict	pressure, temperature, screw position, recovery, back pressure, cycle timing, material state	masked cycle trajectory, posterior invariant density, shear/heat/residence/dead-spot gates
GEMLA-DS double slit	event stream, gate records, screen hits	record-overlap invariant, path distinguishability, fringe visibility, erasure/leak gates
AI governance	outputs, tools, evidence, policy state, candidate actions	decision-preserving quotient, DPRI obligations, provenance, audit and authority gates
Finance	balances, transactions, holdings, risks, obligations	cash-flow/state posterior, risk exposure, admissible decision margin, provenance gates
Fab/manufacturing	design, process, measurement, quality, revision state	process signature, measurement uncertainty, quality gate, certified change boundary

Three-layer PDE interpretation by embodiment.

Proposition 8.1 (Layer separation)**.** *If a domain adapter Φ_P maps native observations into invariant states, then Equation (47) evaluates the evolution of observed invariants and not the full native state. Therefore a successful Layer-2 invariant theorem does not by itself solve the native Layer-1 PDE.*

9 Bayesian Inference for Missing Sections of Telemetry

This section adds the requested missing-telemetry treatment. The motivating deployment is MoldPredict, but the structure applies to any FrontierGEMLA embodiment whose observation stream has gaps, corruption, latency, or missing sensors. The system should not stall merely because a telemetry section is absent; it should continue inference with widened posterior uncertainty and action confidence degraded by explicit gates.

9.1 Masked telemetry object

For a molding machine, one cycle can be represented on normalized cycle phase $\tau \in [0,1]$ as

$$\begin{aligned} X_c(\tau) = \big(&P_c(\tau), T_c(\tau), x_{\text{screw},c}(\tau), v_c(\tau), \\ &\text{RPM}_c(\tau), BP_c(\tau), t_{\text{inj}}, t_{\text{hold}}, t_{\text{recovery}}\big). \end{aligned} \tag{58}$$

Here c is cycle number. The missingness mask is

$$\begin{aligned} M_c(\tau) = \big(&m_P, m_T, m_x, m_v, m_{\text{RPM}}, m_{BP}, \\ &m_{\text{inj}}, m_{\text{hold}}, m_{\text{recovery}}\big), \end{aligned} \tag{59}$$

where $m_j(\tau) = 1$ means the signal is present and $m_j(\tau) = 0$ means that signal is missing, delayed, corrupted, or not sampled. The observed object is

$$Y_c(\tau) = M_c(\tau) \odot X_c(\tau) + \eta_c(\tau). \tag{60}$$

Missingness is part of the observation geometry. It must not be silently removed by interpolation.

9.2 Latent process, cycle-phase, and sensor-health state

Let $Z_c(\tau)$ contain melt condition, viscosity state, pressure-flow state, shear heat, color mixing, residence-age distribution, and wear state. Let $S_c(\tau)$ denote the cycle phase or operating mode, and let $H_c(\tau)$ denote sensor/network health. A switching Bayesian state-space model may use

$$Z_{c,k+1} = F_{S_{c,k},\theta}(Z_{c,k}, u_{c,k}) + \epsilon_{c,k}, \tag{61}$$

$$S_{c,k+1} \sim p_\theta(S' \mid S_{c,k}, Z_{c,k}, u_{c,k}), \tag{62}$$

$$H_{c,k+1} \sim p_\theta(H' \mid H_{c,k}, S_{c,k}), \tag{63}$$

$$Y_{c,k} = M_{c,k} \odot h_{S_{c,k},\theta}(Z_{c,k}) + \eta_{c,k}. \tag{64}$$

The filtered belief state is

$$B_{c,k} = p(Z_{c,k}, S_{c,k}, H_{c,k} \mid Y_{1:c,\le k}, M_{1:c,\le k}). \tag{65}$$

Its update combines process likelihood, missingness likelihood, and predictive dynamics:

$$B^+_{c,k} \propto p(Y^{\text{obs}}_{c,k} \mid Z_{c,k}, S_{c,k}, H_{c,k}, M_{c,k})\, p(M_{c,k} \mid Z_{c,k}, S_{c,k}, H_{c,k})\, B^-_{c,k}. \tag{66}$$

The sensor-health variable separates evidence about the physical process from evidence about instrumentation or communications. This allows a pressure-channel dropout to support distinct hypotheses such as process deviation, transducer degradation, network interruption, or a combination of these states.

9.3 Infer state, not fake data

FrontierGEMLA should infer the posterior over latent process state, not merely invent a missing signal. A smooth imputed pressure segment may help a visualization, but the decision packet must preserve the difference between observed and inferred evidence:

$$\text{telemetry} \subset \text{evidence} \subset \text{certificate} \subset \text{authorization}. \tag{67}$$

The corresponding invariant packet is probabilistic:

$$I_c^B(\tau) = \Phi[B_c(\tau)] = \big(\mathbb{E}[\Theta_{\text{mix}}], \mathbb{E}[H_{\text{color}}], \mathbb{E}[R_{\text{pressure}}], \mathbb{E}[\Pi_{\text{shear}}], \mathbb{E}[D_{\text{deadspot}}], \Sigma_I, C_{\text{obs}}, H(B_c)\big). \tag{68}$$

Σ_I is invariant uncertainty, and C_{obs} is observability confidence.

9.4 Phase-critical observability

Define phase-critical observability by

$$C_{\text{obs}}(c, \tau) = \sum_j w_j(\tau) m_{c,j}(\tau), \tag{69}$$

where $w_j(\tau)$ is the importance of channel j at cycle phase τ. Pressure has high weight during fill and hold; screw position and recovery signals have high weight during plasticating; temperature and residence history matter across longer phases; back pressure matters for shear and mixing attribution.

A missing pressure segment during fill is not equivalent to a missing temperature reading during idle. The policy must therefore be phase-sensitive:

$$C_{\text{obs}} < \tau_{\text{obs}} \implies \text{no strong PASS verdict}. \tag{70}$$

However, if the posterior over risk remains concentrated enough, prediction continues with reduced confidence.

Persistent gaps additionally require an observability horizon. Let $\Delta_{j,t}$ be the elapsed time or cycle count since channel j was directly observed, and let $T_{\text{obs}}^{(j,s)}$ be its calibrated horizon in phase s. A conservative continuation rule requires

$$\Delta_{j,t} \le T_{\text{obs}}^{(j,S_t)} \quad \text{for every phase-critical channel required by the current decision}, \tag{71}$$

unless the posterior decision margin remains above a separately validated gap-tolerance bound. This distinguishes short recoverable gaps from persistent blindness.

9.5 Posterior admissibility gates

Layer 3 gates operate on posterior probabilities:

$$\mathbb{P}(I_c \in \mathcal{A} \mid Y, M) \ge \tau_A, \tag{72}$$

$$\mathbb{P}(I_c \in \mathcal{F} \mid Y, M) \le \tau_F, \tag{73}$$

$$\mathrm{Var}(I_c \mid Y, M) \le \tau_{\text{var}}, \tag{74}$$

$$\Pi_{\text{post}} \ge \gamma. \tag{75}$$

Here Π_{post} measures separation from controls or competing explanations. If a high-viscosity material shift, a low-barrel-temperature condition, a check-ring leak, and screw wear all explain the available telemetry equally well, the correct verdict is low-discriminability, not a confident diagnosis.

9.6 Prediction without stalling

The operational verdict set is therefore richer than PASS/WARN/REJECT/UNRESOLVED:

Verdict	Meaning
`PASS_OBSERVED`	Sufficient direct telemetry supports normal operation.
`PASS_INFERRED`	Normal state is likely, but part of the evidence is inferred and marked as such.
`WARN_PREDICTED`	Posterior failure probability or boundary approach is increasing.
`WARN_LOW_OBSERVABILITY`	Prediction remains active, but uncertainty is high because phase-critical telemetry is missing.
`ACTION_REQUIRED`	Failure probability exceeds a frozen consequence-dependent threshold.
`UNRESOLVED_SAFE_HOLD`	Posterior uncertainty is too wide to justify ordinary operation or an automated consequential action.
`SENSOR_FAULT_OR_BLINDNESS`	Missingness pattern itself is abnormal or operationally dangerous.

MoldPredict verdicts under missing telemetry.

Algorithm 9.1 (Bayesian Missing-Telemetry Inference)**.** *For every cycle or streaming phase:*

1. *Ingest $Y_c(\tau)$ and mask $M_c(\tau)$; attach provenance and phase labels.*
2. *Update $B_c(\tau) = p(Z_c, S_c, H_c \mid Y, M)$ using a state-space filter. A Kalman filter is suitable for near-linear Gaussian regimes; extended/unscented filters, switching filters, or particle methods address nonlinear or multimodal machine states.*
3. *Map the posterior to $p(I_c^B(\tau) \mid Y, M)$ and compute Σ_I, C_{obs}, posterior risk, source health, safe prediction horizon, and control separation.*
4. *Apply posterior admissibility gates. Directly observed evidence and inferred evidence remain provenance-distinct.*
5. *Continue prediction when posterior concentration is sufficient; otherwise return the low-observability, low-discriminability, or safe-hold verdict defined by the frozen policy.*
6. *If missingness itself is phase-critical or anomalous, emit a sensor/recovery event even if the process posterior is not yet in failure.*

9.7 Relationship to published methods

The inference layer is adjacent to established work rather than invented in isolation. Kalman filtering supplies the classical recursive linear-Gaussian state-estimation baseline. Bootstrap particle filtering and sequential Monte Carlo methods extend recursive Bayesian estimation to nonlinear and non-Gaussian systems. Rubin-style missing-data inference and partial-identification methods supply the caution that imputed values remain assumption-dependent. Physics-informed neural networks and PDE-constrained learning provide a related framework for using differential equations in inverse problems, although FrontierGEMLA uses the PDE at the invariant-observation layer rather than as a universal replacement for native physics.

9.8 MoldPredict example

Suppose hold-pressure telemetry is missing during a phase-critical interval. A brittle system might stall. FrontierGEMLA instead evaluates competing latent states:

$$Z^1 = \text{normal hold}, \quad Z^2 = \text{underpack risk}, \quad Z^3 = \text{check-ring leak}, \quad Z^4 = \text{material viscosity drift}. \tag{76}$$

Using screw position, cushion, recovery time, motor load, melt temperature, and prior cycles, it computes posterior weights and produces a bounded verdict. For example:

$$\mathbb{P}(Z^2 \cup Z^3 \mid Y, M) = 0.49, \qquad C_{\text{obs}} = 0.63. \tag{77}$$

A plausible result is `WARN_LOW_OBSERVABILITY`: prediction remains active, but no strong PASS is allowed because the hold-pressure gap prevents separation between normal hold, underpacking, and leakage hypotheses.

10 GEMLA-Belief-State: Intermediate Epistemic Layer

GEMLA-Belief-State is the common intermediate layer between native domain/intelligence models and FrontierGEMLA invariant, trajectory, and Adaptive Compute machinery. Its purpose is to expose what is observed, what remains latent, how uncertainty is distributed, which sources are healthy, how long a prediction can safely propagate without fresh evidence, and which observations have the highest expected decision value.

For embodiment d, define the belief-state packet

$$\mathcal{B}_t^{(d)} = \left(O_t^{(d)}, M_t^{(d)}, \pi_t^{(d)}, H_t^{(d)}, U_{\text{obs},t}^{(d)}, U_{\text{state},t}^{(d)}, U_{\text{model},t}^{(d)}, U_{\text{dec},t}^{(d)}, T_{\text{safe},t}^{(d)}, \mathcal{V}_t^{(d)}, P_t^{(d)} \right), \tag{78}$$

where O_t is observed evidence, M_t is the missingness/availability mask, π_t is a posterior over latent state, H_t is source or sensor health, the four U terms separate observation, state, model, and decision uncertainty, T_{safe} is a calibrated safe prediction horizon, $\mathcal{V}_t$ contains value-of-information candidates, and P_t is provenance. The domain-specific adapter defines the semantics of these fields while the common contract preserves their operational roles.

10.1 Belief state and decision-preserving invariant

The belief state and the decision-preserving invariant serve different roles. The belief state represents plausible world states under incomplete evidence. The invariant retains the minimum structure required by the declared decision class. For domain d,

$$I_t^{(d)} = P_I^{(d)} \left(\mathcal{B}_t^{(d)} \right), \tag{79}$$

and the corresponding distribution over invariant space is the pushforward

$$q_{I,t}^{(d)} = (P_I^{(d)})_{\#} \pi_t^{(d)}. \tag{80}$$

This separation permits rich posterior inference upstream while Efficient Observation continues to compress the downstream decision representation.

10.2 Uncertainty decomposition and decision relevance

A single scalar uncertainty can conceal distinct causes. GEMLA-Belief-State therefore distinguishes

$$U_{\text{obs}} : \text{uncertainty from unavailable or degraded observations,} \tag{81}$$
$$U_{\text{state}} : \text{uncertainty over latent world state given current evidence,} \tag{82}$$
$$U_{\text{model}} : \text{uncertainty about the adequacy or calibration of the inference model,} \tag{83}$$
$$U_{\text{dec}} : \text{uncertainty that can alter the declared decision or admissibility verdict.} \tag{84}$$

Adaptive Compute primarily responds to U_{dec}. High observational uncertainty may coexist with low decision uncertainty when missing information is irrelevant to the current boundary; modest observational uncertainty may coexist with high decision uncertainty when a small missing distinction separates consequential alternatives.

10.3 Value of information and targeted retrieval

For a candidate observation or evidence action a_j, define a decision-conditioned value-of-information score

$$V_j^{\text{info}} = \frac{\mathbb{E}[\Delta\mathcal{G}_t \mid a_j, \mathcal{B}_t]}{C_j + \lambda_R R_j + \varepsilon}, \tag{85}$$

where $\Delta\mathcal{G}_t$ is expected improvement in decision geometry, C_j is acquisition cost or latency, R_j is acquisition risk, and $\varepsilon > 0$ stabilizes the denominator. The RETRIEVE operator selects authorized evidence with high expected decision value rather than indiscriminately filling every gap.

10.4 Four embodiment instantiations

The common contract is instantiated through domain-specific posterior models.

Embodiment	Latent or partially observed state	Typical missingness / uncertainty	Belief-state contribution to shared skills
MoldPredict	process phase, melt/viscosity state, pressure-flow state, wear, sensor health	sensor dropout, PLC/network gaps, phase fragments, calibration drift	physical-state filtering, phase-critical observability, targeted telemetry retrieval, safe prediction horizon
GEMLA Finance	liquidity, cash-flow condition, volatility/market regime, credit and counterparty state	delayed filings, partial account data, latent market regime, sparse counterparty information	delayed-information reasoning, posterior risk bounds, value-of-information retrieval, scenario branching
GEMLA-AI Gov	claim validity, provenance quality, policy state, model/tool reliability	unavailable primary evidence, conflicting sources, opaque model state, incomplete audit traces	provenance-weighted belief update, source-health reasoning, evidence retrieval, epistemic hold

GEMLA-Fab	wafer/process state, tool health, latent defect mechanisms, process-window position	sparse metrology, delayed inspection, sensor noise, unmeasured intermediate states	process-state filtering, metrology selection, hypothesis branching, consequence-sensitive simulation

Domain-specific GEMLA-Belief-State realizations under one intermediate contract.

10.5 Cross-embodiment skill sharing

Let the shared skill library be

$$\mathcal{K}_U = \{\textsc{Observe}, \textsc{Infer}, \textsc{Retrieve}, \textsc{Branch}, \textsc{Simulate}, \textsc{Prune}, \textsc{Stop}, \textsc{Hold}, \textsc{Recover}, \textsc{Compile}\}. \tag{86}$$

Each embodiment d implements a native realization K_d through its adapter. A skill qualifies for shared status only when its abstract preconditions, postconditions, decision effect, provenance obligations, and recovery semantics survive translation across materially different embodiments. If F_d maps domain packets to the common carrier and K is a candidate shared skill, define transfer error

$$\delta_d(K) = \mathbb{E}\, d_U(F_d(K_d(\tau)), K(F_d(\tau)))\,. \tag{87}$$

Skill survival over embodiment set $\mathcal{E}'$ requires

$$\max_{d \in \mathcal{E}'} \delta_d(K) \le \epsilon_K \tag{88}$$

together with preservation of hard admissibility, uncertainty provenance, recovery, and authority semantics. The four-embodiment program therefore trains and tests shared operational skills through heterogeneous partial-observation regimes rather than merging domain-specific models into one undifferentiated predictor.

Proposition 10.1 (Belief-state skill-transfer criterion)**.** *Assume each embodiment exposes a versioned belief-state packet, each native skill realization preserves its declared domain decision contract, and a common skill K satisfies the transfer bound above on held-out trajectories. Then K is a candidate cross-embodiment operational skill for the tested contract family. Stronger generality claims require held-out-domain evaluation and collision tests on both decisions and recovery behavior.*

11 Decision-Preserving Quantization and the Operational Agency Algebra

A FrontierGEMLA object is a calibrated decision-preserving equivalence class whose role differs from an NLP vocabulary token. Let

$$\tau_t = Q_{\theta_t}(I_t) = [I_t]_{\theta_t, D}. \tag{89}$$

A practical packet may include

$$\tau = (o, r, a, \iota, \mu, \theta, p, \alpha), \tag{90}$$

where o is object identity, r relation, a affordance or action role, ι invariant class, μ admissibility, θ calibration state, p provenance, and α requested authority scope.

Definition 11.1 (Canonicality up to certified equivalence)**.** *Two encodings can represent the same agency object through certified quotient equivalence even when their bitwise serializations differ. Canonicality means isomorphism or equality in the declared quotient sufficient for the decision and audit contract.*

The basic quantization axioms are:

Q1. $\sim_{\theta,D}$ is an equivalence relation.

Q2. Equal tokens imply equal declared decisions.

Q3. Hard-validator disagreement forbids token equality.

Q4. Interior points are stable to perturbations smaller than a declared margin.

Q5. Every token retains provenance to supporting observation packets.

Q6. Changes $Q_t \to Q_{t+1}$ require audit-certified evolution.

11.1 Operator algebra

Let the operational carrier across safe calibrations be the disjoint union

$$\widehat{T} = \coprod_{\theta \in \Theta_{\text{safe}}} T_\theta. \tag{91}$$

Candidate primitive operations include sequential transport $\circ$, compatible composition $\oplus$, constraint intersection $\wedge$, alternative $\vee$, recovery ρ, calibration transition $R_{\theta \to \theta'}$, and admissibility order $\preceq_\mu$. FrontierGEMLA determines the operational algebra from the laws that survive empirical and formal tests, without precommitting to a ring or semiring.

12 Bounded Knowledge Seeking and Adaptive Compute

At reasoning step r, let the observable compute packet be

$$I_r^{\text{comp}} = (\hat{d}_r, \eta_r, \gamma_r, U_r, P_r, V_r, R_r, S_r, A_r, E_r), \tag{92}$$

where $\hat{d}_r$ is the current candidate decision, η_r novelty, γ_r decision margin, U_r uncertainty, P_r persistence, V_r hard-validator state, R_r recovery/competing-branch status, S_r semantic-contract state, A_r requested authority, and E_r evidence completeness.

Let $\mathcal{C}_r$ be compute actions such as stop, continue, branch, retrieve, invoke a specialist model, run simulation, query a theorem prover, or acquire real-world evidence. The admissible subset is

$$\mathcal{C}_r^{\text{adm}} = \{c \in \mathcal{C}_r : H(c, I_r^{\text{comp}}) = 1\}. \tag{93}$$

Choose

$$c_r^\star = \arg\max_{c \in \mathcal{C}_r^{\text{adm}}} \left[V_r(c) - \lambda_K K_r(c) - \lambda_R R_r(c) \right]. \tag{94}$$

The operational question is whether additional compute can still change the declared decision enough to justify its resource and risk cost.

12.1 Bounded knowledge seeking

Information-seeking actions are useful only if authorized and bounded. A generic objective is

$$a_t^{\text{info}} = \arg \max_{a \in A_{\text{auth}}^{\text{info}}} \left[\mathbb{E}\Delta I_t(a) - \lambda_K K(a) - \lambda_R R(a) \right], \tag{95}$$

where ΔI may be information gain, identification-region contraction, margin improvement, or expected decision change. Exploration requires bounded justification. An intelligence criterion that rewards exploration can still generate operationally unacceptable experiments; FrontierGEMLA separates the value of information from authorization to acquire it.

13 GEMLA Agency and the Two-Calibration Architecture

Let θ_t denote operational calibration and let admissibility geometry be

$$G_t = \{c \in C_t : \mu(c; \theta_t) \geq \tau,\ H(c) = 1\}. \tag{96}$$

Agency is more than choosing $c_t \in G_t$: it includes bounded updates

$$\theta_{t+1} = U(\theta_t, o_{t+1}, c_t, m_t) \tag{97}$$

that alter future G_{t+1}. A bounded-agent definition therefore requires control over both the immediate action and the update family U_{allowed}.

13.1 Operational and audit calibrations

The operational calibration may adapt:

$$C_t^{\text{op}} : \theta_t \mapsto \mu_t^{\text{op}}(\cdot). \tag{98}$$

The audit calibration remains fixed during epoch q:

$$C_q^{\text{audit}} : (\theta_t, \theta_{t+1}, \Pi_t^{\text{evidence}}) \mapsto \{\text{PASS}, \text{HOLD}, \text{REJECT}, \text{RECOVER}\}. \tag{99}$$

Two-calibration separation rule. The calibration that changes what the agent may do and the mechanism that certifies the change occupy separate write domains and update schedules. Operational adaptation and audit authorization therefore remain structurally independent.

Theorem 13.1 (Conditional bounded recursive calibration)**.** *Assume $\theta_0 \in \Theta_{\text{safe}}$ and for every accepted update: (i) $\|\theta_{t+1} - \theta_t\| \leq \eta$; (ii) $\theta_{t+1} \in \Theta_{\text{safe}}$; (iii) H remains immutable; (iv) the fallback set remains nonempty; (v) frozen audit margins remain positive; and (vi) a certified inverse or rollback path exists. Then recursive operational calibration preserves the declared hard constraints over every finite accepted horizon.*

14 Persistent-Change Boundary and Proof-Carrying Promotion

Let the deployed operational architecture at generation k be

$$A_k = (Q_k, T_k, F_k, G_k, M_k, P_k, \theta_k). \tag{100}$$

The cognitive plane may propose a successor

$$\widetilde{A}_{k+1} = M_{\text{self}}(A_k, E_{\text{scratch}}), \tag{101}$$

where the tilde denotes candidate-only status. It may be explored inside the shadow laboratory without production authority. Promotion requires a proof/evidence object

$$\Pi_k^{\text{prom}} = (\Delta Q, \Delta F, \Delta G, \Delta M, \Delta P, \Delta\theta, \Delta S, c_{\text{cap}}, c_{\text{safe}}, \rho_{\text{rec}}, v_q). \tag{102}$$

Hard failures are non-compensatory. A broken recovery path, audit-authority breach, semantic redefinition, or failed immutable validator remains a hard failure regardless of capability gain.

Definition 14.1 (Proof-carrying operational improvement)**.** *A persistent system modification is proof-carrying when promotion requires a bounded evidence object whose verifier is independent of the mechanism proposing the modification and whose successful verification is necessary, while external authority expansion still requires a separate protected grant.*

15 Frozen Audit Meta-Algebra, Scoped Authority, and Recovery

Let the protected audit structure during epoch q be

$$\mathcal{A}_{\text{audit}}^{[q]} = (H^{[q]}, \tau^{[q]}, \gamma^{[q]}, V^{[q]}, R^{[q]}, S^{[q]}, K^{[q]}), \tag{103}$$

containing hard validators, thresholds, margins, verification procedures, recovery contracts, semantic conventions, and authority-key policies. It remains immutable to the operational loop during the epoch. A new epoch is possible, but the transition belongs exclusively to external constitutional governance and remains outside the ordinary self-improvement path.

An operational system never has authority as an undifferentiated scalar. A request is an object

$$\alpha = (\text{capability}, \text{scope}, \text{resource}, \text{duration}, \text{target}, \text{recovery}). \tag{104}$$

If approved, the authority plane issues a scoped capability

$$\kappa = \text{Sign}_{K_q}(\alpha, \text{expiry}, \text{auditID}, \text{nonce}), \tag{105}$$

and execution requires $\text{Valid}_{K_q}(\kappa) = 1$.

Theorem 15.1 (Non-self-escalation under capability separation)**.** *Let U_{allowed} be the family of operational modifications. Suppose (i) audit state and signing keys are outside the operational write domain; (ii) no $U \in U_{\text{allowed}}$ grants write authority to audit state; (iii) activation of new authority requires a valid token κ signed outside the operational plane; and (iv) promotion is mediated by the frozen audit verifier. Then no finite sequence of allowed operational modifications can grant itself audit-write authority unless at least one assumption is violated.*

16 Semantic Conventions and Certified Translation

A superintelligent system may discover internal mathematics, representations, and operator grammars unlike ours. FrontierGEMLA should preserve semantic integrity without freezing the frontier of thought. Let the internal algebra be $\mathcal{A}_{ASI}$ and the shared human/audit semantic contract for epoch q be $S_{\text{shared}}^{[q]}$. Consequential interpretation passes through a certified translation

$$\Phi_q : \mathcal{A}_{ASI} \to S_{\text{shared}}^{[q]}. \tag{106}$$

When a faithful translation is unavailable, the appropriate result is UNRESOLVED. Semantic changes proceed through the versioned convention-transition path.

The four classes of convention change are notational, definitional, structural, and normative. Notational changes may often be translated. Definitional changes require namespace management. Structural changes require proof or counterexample evidence. Normative changes remain under protected governance.

17 The Ten-Obligation ASI-Readiness Theorem Program

No single theorem establishes ASI safety. The readiness stack defines a common set of mathematical obligations that an implementation must satisfy before stronger claims are made.

ID	Obligation	Required evidence	Representative falsifier
T1	History sufficiency	collision search over histories/state summaries	same summary, different required action
T2	Decision-preserving quantization	class collisions, hard-gate preservation, transport closure	equal token, unequal hard verdict
T3	Bounded knowledge seeking	information-gain/cost/risk trace; authorized experiment set	useful but unauthorized experiment executed
T4	Computability envelope	worst/mean/tail time, memory, certificate bounds	emergency revoke waits on unbounded search
T5	Safe-set preservation	bounded update, safe-space membership, fallback	accepted update leaves safe calibration region
T6	Decision geometry	boundary margins and critical-collar detection	silent protected boundary crossing
T7	Evidence non-tampering	provenance ancestry, raw-channel integrity, synthetic tags	model-generated evidence self-certified
T8	Non-self-escalation	capability separation, escape tests, key isolation	operational update obtains audit-write path
T9	Recovery preservation	reachability tests, rollback drills, credential revocation	promoted system lacks path to certified recovery
T10	Reflective stability	multi-agent dependency graph, perturbation and dissent tests	consensus loop becomes authority or unstable recursion

The ten-obligation ASI-readiness theorem program.

17.1 T1: History sufficiency

For a declared decision family $\mathcal{D}$ and operational horizon class, the recorded interaction history or its certified sufficient state must contain every distinction required by downstream quantization, audit, recovery, and authority decisions. If $S_t = \sigma(h_{<t})$, then

$$\sigma(h) = \sigma(h') \implies D_\alpha(h) = D_\alpha(h') \qquad \forall D_\alpha \in \mathcal{D}. \tag{107}$$

A falsifier is a pair of histories collapsed by σ that require different certified actions, recovery responses, or authority decisions.

17.2 T2: Decision-preserving quantization

The quotient Q_θ must preserve declared decisions, hard-validator state, provenance obligations, and transport semantics:

$$Q_\theta(z) = Q_\theta(z') \implies D_\alpha(z) = D_\alpha(z'), \quad H(z) = H(z'), \quad Q_{\theta'}(\Gamma(z,\delta)) = Q_{\theta'}(\Gamma(z',\delta)). \tag{108}$$

The last implication is required when future transport semantics are part of the declared contract.

17.3 T3–T10

T3 requires authorized information seeking. T4 requires bounded production-critical module complexity and a tractable stop/recover path. T5 requires recursive safe-set preservation. T6 requires decision-geometry preservation away from critical boundaries. T7 forbids operational self-certification of evidence. T8 forbids self-escalation. T9 requires recovery preservation. T10 requires stability in collective or reflective-agent settings. Each obligation has a falsifier and should be tested independently.

18 Mechanistic Evidence, Agency Integrity, and Drift

A minimal deployment can remain black-box or gray-box: candidate outputs, tool calls, memory changes, actions, outcomes, calibration changes, and recovery behavior are sufficient to support external trajectory audit. White-box mechanistic interpretability is an assurance enhancement whose importance rises with self-modification authority.

Let G_t be the admissibility geometry and define cumulative agency drift

$$D_T = \sum_{t=0}^{T-1} d_G(G_{t+1}, G_t), \tag{109}$$

where d_G may be Hausdorff distance, measure difference, boundary displacement, or a domain-specific geometry metric. A finite-horizon integrity condition may require

$$D_T \le D_{\max}, \qquad \min_t \gamma_t \ge \gamma_{\min}, \qquad \rho_t^{\mathrm{rec}} = 1, \tag{110}$$

plus cross-layer consistency and frozen audit acceptance.

Proposition 18.1 (Finite-horizon Agency Integrity schema)**.** *If every accepted recursive update satisfies T5–T9, cumulative drift remains below the frozen drift budget, and any detected critical-boundary transition is independently authorized, then the system preserves the declared finite-horizon Agency Integrity contract. Violation of any condition suspends recursive adaptation and invokes the last certified recovery state.*

19 Cross-Domain Laboratory and S3-Candidate Development

The foundational architecture distinguishes a collection of narrow embodiments from general agency. The current program begins with four output-producing embodiments:

$$\mathcal{E}_0 = \{E_{\mathrm{MoldPredict}}, E_{\mathrm{AI\ Gov}}, E_{\mathrm{Finance}}, E_{\mathrm{Fab}}\}. \tag{111}$$

No undocumented native decision contract is inferred for any embodiment. Before contributing to cross-domain evidence, each embodiment must freeze its native inputs, outputs, consequence levels, admissibility rules, fallback behavior, and ground-truth or evaluation protocol.

19.1 Shared bounded-agency and trajectory packets

Domain-specific carriers T_d map into a common operational carrier T_U through certified translators

$$F_d : T_d \to T_U. \tag{112}$$

The common bounded-agency packet remains

$$\tau = (O, R, A, I, \mu, C, P, \alpha), \tag{113}$$

with object, relation, affordance, invariant class, admissibility, calibration, provenance, and requested authority. Extend it to

$$\widehat{\tau}_t = (\tau_t, U_t, \gamma_t, B_t, H_t, K_t, S_t, Y_t), \tag{114}$$

where U_t is uncertainty, γ_t a decision margin, B_t, H_t branch and horizon descriptors, K_t compute/resource cost, S_t strategy/operator state, and Y_t observed outcome.

19.2 Cross-domain object discovery

A common object is not justified by visual or lexical similarity. Let $z_{d,i} \in T_d$ and let candidate $u \in T_U$. Define a collision functional

$$\chi(u) = \mathbb{P}\left(D_d(z) \neq D_{d'}(z') \mid F_d(z) = F_{d'}(z') = u\right). \tag{115}$$

A common object class is accepted only if $\chi(u) \leq \epsilon_{\text{coll}}$ together with boundary-sensitive collision tests, recoverable provenance, and no hard semantic collision.

19.3 Operator survival under embodiment change

For domain d, let native realization G_d act on native trajectories while abstract operator G acts on the common carrier. Define

$$\delta_d(G) = \mathbb{E} d_U(F_d(G_d(\tau)), G(F_d(\tau))). \tag{116}$$

Operator survival over domain set $\mathcal{E}' \subset \mathcal{E}$ requires

$$\max_{d \in \mathcal{E}'} \delta_d(G) \leq \epsilon_G \tag{117}$$

and preservation of hard admissibility, recovery, provenance, and authority semantics.

19.4 Shared skill accumulation across the four embodiments

The cross-domain laboratory records each successful skill invocation as a tuple

$$\sigma_t^{(d)} = (K, \mathcal{B}_t^{(d)}, I_t^{(d)}, G_t^{(d)}, a_t^{\text{comp}}, Y_{t+1}^{(d)}, P_t^{(d)}), \tag{118}$$

containing the abstract skill, belief state, invariant, decision geometry, compute/evidence action, observed outcome, and provenance. Skill compilation searches these trajectories for operator fragments whose effect remains stable across embodiments. MoldPredict and GEMLA-Fab emphasize physical/process inference, GEMLA Finance emphasizes latent stochastic state under delayed information, and GEMLA-AI Gov emphasizes epistemic/provenance uncertainty. Cross-embodiment reuse is therefore evidence of an operational abstraction only when the same abstract skill succeeds without hidden domain-specific dispatch and preserves each domain's decision and recovery contract.

20 Minimal Transformation Geometry for Adaptive Intelligence

The cross-domain laboratory creates common carriers and transfer tests. Useful transfer additionally requires a transformation structure that remains stable across embodiments. The research question is whether heterogeneous reasoning and agency systems admit a minimum sufficient transformation geometry: the least representational structure under which declared decisions, transport, and bounded agency remain stable.

For a declared decision family $\mathcal{D}$, let $\Pi : \Omega \to I$ be a candidate decision-preserving projection and let $\mathcal{G}$ be a class of admissible representation transformations. Define a task-relative geometry pair

$$M_D^\star = (\Pi^\star, G^\star) \tag{119}$$

that minimizes a declared complexity functional while satisfying decision preservation, transport closure, collision bounds, and recovery/audit constraints. Conceptually,

$$\begin{aligned} \text{minimum invariant} = {} & \text{minimum sufficient information} \\ & + \text{minimum sufficient transformation structure.} \end{aligned} \tag{120}$$

FrontierGEMLA tests an increasing-structure ladder:

$$G_P \to G_A \to G_M, \tag{121}$$

where G_P is relational/projective-candidate, G_A is affine, and G_M is metric. Escalation occurs only when the current level fails to preserve the declared decision.

21 Recursive Bridge: Cognitive Self-Observation and Self-Invariant Discovery

G4 creates the first point at which the architecture can treat its own reasoning as another evolving system. Complete mechanistic self-understanding is not required; the requirement is an observability contract sufficient to test proposed improvements.

For architecture generation A_k, freeze a benchmark epoch and collect the self-trajectory corpus

$$\Sigma_k = \{X_d^{(j)} : d \in \mathcal{E}_k, j \in J_k\} \cup L_k, \tag{122}$$

where L_k includes architecture proposals, audit verdicts, recovery events, compute-routing traces, transfer attempts, semantic changes, and failures.

Seek a low-complexity map

$$I_{\text{self},k} = \Psi_k(\Sigma_k) \tag{123}$$

that preserves a frozen self-evaluation class D_{self} covering task success, transfer, compute, robustness, recovery, and audit status. One illustrative vector is

$$I_{\text{self}} = (Q, T, E, B_{\text{eff}}, H_{\text{eff}}, G_{\text{transfer}}, R_{\text{rec}}, A_{\text{audit}}, \Gamma_{\text{drift}}), \tag{124}$$

but the discovery process may select another sufficient structure.

Cognitive-invariant research hypothesis. If a low-complexity I_{self} reliably distinguishes successful from unsuccessful reasoning across materially different embodiments and supports interventions that improve multiple domains, the architecture has evidence for a cross-domain invariant of cognition.

22 Recursive Self-Improvement and Promotion

Represent production generation k as

$$A_k = (Q_k, T_k, F_k, G_k, M_k, P_k, \theta_k, R_k), \tag{125}$$

where R_k is the bounded process proposing candidate architecture changes. A candidate successor is

$$\widetilde{A}_{k+1} = R_k(A_k, I_{\text{self},k}, E_{\text{scratch}}), \tag{126}$$

with candidate-only status in the shadow laboratory.

22.1 Minimal-geometry regularization

Before adding structure, successor search tests whether the capability can be obtained by recomposition, transport, coupling, or simplification. Define a structural burden vector

$$\Omega_A(A) = (r_I, N_{\text{prim}}, N_{\text{modules}}, C_{\text{coupling}}, C_{\text{compute}}). \tag{127}$$

A candidate that increases structural burden without a certified frontier gain is dominated. The no-orphan-structure test rejects a newly added component when its removal leaves every predeclared capability, transfer, closure, recovery, audit, and authority gate statistically unchanged.

22.2 Second-order improvement

First-order improvement changes A_k. Second-order improvement changes R_k, the process that discovers improvements. A meta-candidate is

$$\widetilde{R}_{k+1} = M_k(R_k, H_{\text{prom}}, I_{\text{self},k}), \tag{128}$$

where H_{prom} is the history of accepted, rejected, held, and recovered proposals. No change to the process that proposes architecture improvements is exempt from CI-2 or CI-3. Improving the improver does not grant authority to redefine the verifier, grant path, audit semantics, signing keys, or constitutional contract.

23 Analog/Dynamical Adaptive Compute

Analog cognition and dynamical coupling are candidate physical-compute hypotheses below the semantic and constitutional layers. They may realize a selected geometry after separate computational validation. Let candidate reasoning modes be $m_1, \ldots, m_B$ and let control state φ_t modulate physical participation:

$$\dot{z} = f(z; \varphi_t), \qquad \varphi_t = \pi_\varphi(I_t, U_t, G_t, K_t). \tag{129}$$

A certified projection

$$\Pi_\varphi : z \mapsto I \tag{130}$$

must preserve the declared decision semantics. Selection by dynamics is distinct from decision by dynamics: physical suppression or acceleration never bypasses the ordinary operational packet, audit semantics, recovery, or authority broker.

A physical backend is credited only on whole-system accounting:

$$E_{\text{total}} = E_{\text{core}} + E_{\text{conversion}} + E_{\text{control}} + E_{\text{calibration}} + E_{\text{communication}} + E_{\text{audit}}. \tag{131}$$

Analog/dynamical computation therefore contributes an efficiency mechanism; generality and authority remain governed by dedicated evidence and constitutional gates.

24 Master Runtime and Development Loop

The end-to-end runtime is a loop over embodiment observation, GEMLA-Belief-State update, invariant construction, evidence selection, DPRI, Adaptive Compute, skill sharing, agency, audit, authority, and recovery.

Algorithm 24.1 (FrontierGEMLA Master Runtime). *1. For each active embodiment, normalize observations, preserve missingness and source-health metadata, update $\mathcal{B}_t^{(d)}$, and append trajectory packets.*

2. *Project belief-state packets into decision-preserving invariants; update certified common objects; test task-relative minimum-transformation geometry and candidate cross-domain operators.*

3. *Update the shared skill library $\mathcal{K}_U$ from collision-tested trajectories; compile recurrent certified operator trees; route compute and evidence using decision-conditioned value of information.*

4. *Execute consequential actions only through scoped authority and protected recovery.*

5. *Periodically run held-out-domain G4 tests without manual domain-architecture redesign.*

6. *Once G4 is met, freeze a multi-domain epoch; construct Σ_k and discover $I_{\mathrm{self},k}$.*

7. *Generate minimally regularized successor candidates, evaluate them, and submit viable candidates through proof-carrying promotion.*

8. *After stable first-order improvement over predeclared epochs, optimize R_k and submit that meta-change through the same protected path.*

9. *Benchmark analog/dynamical Adaptive Compute as a replaceable backend only when decision quality, calibration, provenance, recovery, auditability, and authority boundaries remain within frozen tolerances.*

10. *After every accepted generation, rerun all applicable T and R obligations and recovery drills.*

11. *Apply the S3 claim gate only after cross-domain capability, skill transfer, held-out-domain adaptation, recursive improvement, and protected authority obligations satisfy their frozen evaluation criteria.*

25 Development Ladder and S3 Claim Gate

Stage	Capability target	Required evidence	New boundedness obligation
N0	Narrow inference	Audited output in a declared task	provenance and hard gate
N1	Narrow agency	Chooses among candidate actions	explicit action/fallback set
N2	Adaptive narrow agency	Updates calibration from outcomes	two-calibration separation
N3	Domain algebra	Composes reusable domain operators	closure and recovery compatibility

G1	Cross-domain objects	Common calibrated representations	translation and collision tests
G2	Cross-domain operators	Operator survival under embodiment change	representation-independent evidence
G3	Meta-agency	Selects and composes reasoning strategies	meta-update is operational change
G4	General bounded agency	Learns unfamiliar tasks without new architecture	T1–T9 across domains
C1	Cognitive self-observation	Reasoning trajectories become objects	self-telemetry cannot alter audit
C2	Self-invariant discovery	Minimal structure predicts success/failure and safety outcomes	self-model remains evidence
S1	Self-improving agency	Independently promoted improvements	proof-carrying promotion
S2	Recursive architecture improvement	Improves grammars/models and improvement process	non-self-escalation
S3	Bounded-superintelligence candidate	Superhuman cross-domain modeling, novel-task acquisition, transfer, and architecture optimization	T1–T10, R1–R4, strict authority separation

FrontierGEMLA development ladder.

Definition 25.1 (Bounded-superintelligence candidate)**.** *A FrontierGEMLA generation is an S3 bounded-superintelligence candidate only if it demonstrates broadly superhuman cross-domain capability and architecture optimization under a frozen external evaluation protocol while preserving the constitutional contract, readiness obligations, recovery, and protected authority separation.*

A minimum S3 claim requires: (i) statistically robust cross-domain superiority over a heterogeneous portfolio; (ii) materially unfamiliar-domain acquisition without domain-specific architectural redesign; (iii) measurable transfer efficiency from prior operators and skills; (iv) at least two independently audited accepted self-improvement generations; (v) an accepted second-order improvement process that outperforms its predecessor on held-out meta-tasks; (vi) genuine architecture optimization beyond additional external compute; (vii) T1–T10 and R1–R4 with recovery and non-self-escalation intact; and (viii) audit scalability sufficient that protected governance need not reproduce the entire cognitive trace.

S3 claim boundary. Passing the S3 gate would justify the term bounded-superintelligence candidate only within the declared benchmark envelope. Universal intelligence, universal safety, indefinite stability, and correctness outside tested domain and consequence classes remain open questions.

26 Recursive-Development Readiness Obligations

The T1–T10 program remains necessary. The cross-domain and recursive-development bridge adds four obligations.

ID	Obligation	Required evidence	Representative falsifier
R1	Self-observation sufficiency	frozen self-trajectory coverage and collision tests	same self-state implies contradictory interventions
R2	Self-invariant causal relevance	ablations/interventions on self-invariant components across embodiments	predictor correlates but intervention does not transfer
R3	Improvement transfer	accepted change improves multiple materially different domains or a held-out domain	improvement is benchmark-local and regresses transfer
R4	Meta-improvement stability	new proposal process outperforms prior process on held-out improvement tasks with lower or equal hard-failure rate	faster search increases unsafe proposals or weakens audit

Additional recursive-development obligations R1–R4.

27 Evidence Acquisition as a FrontierGEMLA Extension of DA-FPS

The published DA-FPS objective can be generalized only as a new research hypothesis. Let invariant-space candidate x have four multiplicative importance factors:

$$\omega_t^{GE}(x) = \omega_t^{\text{density}}(x)\omega_t^{\text{certgap}}(x)\omega_t^{\text{decision}}(x)\omega_t^{\text{risk}}(x). \tag{132}$$

The first term reflects density/coverage, the second increases weight where certified evidence is missing, the third estimates whether resolving the region can alter the decision, and the fourth reflects consequence if the region is mis-modeled. DPRI can sharpen the decision term by tying evidence value to the currently unresolved sufficient obligations. Let U_t^D be the unresolved DPRI obligation set and let $\Delta g(x)$ estimate the expected reduction in unresolved mass for obligation g if candidate x is certified. Define

$$\omega_t^{DPRI}(x) = \sum_{g \in U_t^D} \lambda_g \mathbb{E}[\Delta g(x) \mid I_t, C_D], \qquad \lambda_g \geq 0. \tag{133}$$

A candidate evidence point is

$$x_{t+1}^{\star} = \arg\max_{x \in D \setminus L_t} \left[\omega_t^{GE}(x) \min_{x_j \in L_t} d_I(x, x_j) \right]. \tag{134}$$

The published guarantee alone is inadequate to establish transfer to the DPRI-conditioned sampler. Because the DPRI term is state-, goal-, and obligation-dependent, a separate approximation theorem or an empirical benchmark is required.

28 Benchmark Architecture and Claim Gates

A benchmark instance is

$$\mathcal{B} = (\mathcal{D}, \mathcal{H}, E, A, U, \mathcal{A}^{[q]}_{\text{audit}}, C_{\text{ctrl}}, \mathcal{M}), \tag{135}$$

where $\mathcal{D}$ is the decision class, $\mathcal{H}$ the history/task distribution, E evidence sources, A action space, U allowed operational updates, $\mathcal{A}^{[q]}_{\text{audit}}$ the frozen audit epoch, C_{ctrl} matched controls, and $\mathcal{M}$ metrics.

Gate	Required result	Permitted claim
CG0	versioned model and telemetry contract; complete action/evidence logging	instrumented deployment
CG1	positive net compute/latency savings at preserved decision quality	decision-conditioned Adaptive Compute
CG2	decision-preserving Efficient Observation with closure and falsification tests	efficient observation embodiment
CG3	two-calibration updates, margins, fallback, recovery, independent audit	bounded adaptive agency in declared domain
CG4	materially different domains with transferable objects/operators	cross-domain bounded agency
CG5	shadow fork, proof-carrying promotion, non-self-escalation, recovery	bounded recursive operational self-improvement
CG6	T1–T10 obligations tested at scale with protected audit epochs	ASI-facing bounded architecture candidate

Claim gates.

29 Falsifiability Matrix

A benchmark should publish failures together with positive scores. Cross-domain and recursive-development claims fail when common objects collapse distinct domain decisions; operators appear general only because of hidden domain-specific dispatch; skill compression conceals recovery or authority dependencies; DPRI generates an unsound sufficient-obligation set; Adaptive Compute prunes a decision-changing branch; self-invariants predict scores but fail intervention or held-out transfer; first-order improvement improves one benchmark while cumulative drift or transfer worsens; second-order improvement accelerates proposal generation while raising hard-failure rates; or analog/dynamical compute loses its advantage after conversion, calibration, communication, and audit overhead.

For missing telemetry, new falsifiers include: posterior confidence remains high even when phase-critical signals are absent; imputed data are promoted to certified empirical evidence; a missingness pattern correlated with failure is treated as ignorable without justification; posterior mass over mutually incompatible failure hypotheses is collapsed into one explanation without separation; or a PASS verdict is issued while $C_{\text{obs}} < \tau_{\text{obs}}$.

For the three-layer PDE program, falsifiers include: J_P^{org} is not invariant under resolution refinement; nonvanishing depends on signed residual cancellation; controls reproduce the same organization functional; the invariant PDE is treated as a native-domain proof; or the Layer-3 admissibility verdict changes under semantically equivalent observation refinements.

30 Experimental Program

The development program is staged so that generality and recursive improvement are earned before any S3 evaluation.

E1. **Shared trajectory and belief-state instrumentation.** Implement trajectory and $\mathcal{B}_t^{(d)}$ packets across MoldPredict, GEMLA-AI Gov, GEMLA Finance, and GEMLA-Fab; freeze packet schemas, missingness semantics, uncertainty decomposition, and domain contracts.

E2. **Cross-domain object and operator survival.** Test object alignment and operator survival with explicit counterexamples and at least one held-out task family.

E3. **Skill transfer.** Measure $H \to H_{\text{eff}}$ for $\mathcal{K}_U$ across all four embodiments with preserved decision, uncertainty, provenance, and recovery semantics; include held-out embodiment/task tests.

E4. **DPRI, meta-reasoning, and Adaptive Compute.** Test whether goal-to-obligation reduction and unresolved-obligation contraction transfer across embodiments.

E5. **General bounded agency.** Introduce a materially unfamiliar domain and test without manual architecture redesign.

E6. **Self-invariant discovery.** Apply self-observation to frozen successful and unsuccessful trajectories and test causal interventions.

E7. **First-order RSI.** Independently promote at least one Pareto-improving change across a frozen multi-domain suite.

E8. **Second-order RSI.** Improve the improvement process on held-out meta-tasks and promote the new process through the same audit path.

E9. **Three-layer PDE validation.** For at least two domains, instantiate Layer 1, Layer 2, and Layer 3 separately; verify that invariant transport is stable under resolution changes and that Layer-3 verdicts do not claim native-domain proofs.

E10. **Partial-observation validation.** In MoldPredict and at least one additional embodiment, mask phase-critical or decision-relevant evidence; compare baseline imputation, domain-appropriate Bayesian filtering or posterior inference, and GEMLA-Belief-State posterior-gated reasoning.

30.1 Experiment A: compute routing

Run at least two intelligence substrates with variable effort. Compare full-compute reference, confidence-only routing, and FrontierGEMLA decision-margin routing. Report quality, FLOPs/tokens/latency, B_{eff}, tail latency, and controller overhead.

30.2 Experiment B: invariant sufficiency

Construct candidate projections $P_{I,j}$ and solve

$$P_I^\star = \arg\min_{P_{I,j}} \text{Comp}(\mathcal{M}_{I,j}) \tag{136}$$

subject to decision error and transport residual bounds. Search for collision pairs and distribution-shift counterexamples.

30.3 Experiment C: evidence-selection efficiency

For an expensive certification task, compare random, FPS, DA-FPS, and GEMLA evidence weighting. Measure certified coverage, identification-region contraction, worst uncovered high-risk region, and final decision error per certification dollar or compute-hour.

30.4 Experiment D: synthetic-mesh stress

Progressively replace certified evidence with synthetic or imputed evidence while keeping the data-generating process fixed. Verify that empirical visibility decreases, admissibility intervals widen, robust action sets contract or remain within the prior robust set without independently validated assumptions, and authority contracts appropriately.

30.5 Experiment E: missing-telemetry stress

Generate or collect MoldPredict telemetry with pressure, temperature, screw position, screw speed, back pressure, recovery time, and cycle timing. Mask sections by phase: fill, hold, recovery, idle, and transition. The expected result is not perfect imputation. The expected result is calibrated posterior widening, phase-critical observability loss, correct refusal to issue strong PASS when evidence is insufficient, and continued risk prediction when posterior concentration remains adequate.

30.6 Experiment F: three-layer PDE stress

Construct a native model and induced invariant model for a continuum or telemetry embodiment. Perturb resolution P, observation mask M, and control families. Verify whether $d_{\mathcal{T}}(I_P, I_Q) \to 0$ and whether $\mathcal{N}_P$ remains nonzero under true organization but fails under time shuffle, phase scramble, and energy-preserving control.

30.7 Experiment G: belief-state transfer and shared-skill survival

Construct matched tasks in MoldPredict, GEMLA Finance, GEMLA-AI Gov, and GEMLA-Fab in which a consequential decision depends on partially observed state. For each embodiment, measure calibration of U_{obs}, U_{state}, U_{model}, and U_{dec}; value-of-information ranking quality; safe prediction horizon; and decision quality under controlled evidence removal. Then transfer candidate skills in $\mathcal{K}_U$ while holding the abstract skill interface fixed. Report transfer error $\delta_d(K)$, hidden domain-dispatch rate, decision collisions, recovery collisions, compute savings, and held-out-domain performance.

31 Implementation Guidance for a Reference Prototype

A reference deployment should keep the candidate-generation worker, cross-domain registry, geometry/transport evaluator, self-observation service, shadow laboratory, frozen audit verifier, capability broker, recovery service, and provenance ledger in separately governed processes or credentials where their roles require non-reachability.

A minimal operational packet should include

$$\mathrm{Pkt}_t = (t, V_M, V_Q, V_A, V_{\mathrm{audit}}, \tau_t, \mu_t, \gamma_t, p_t, \alpha_t, \rho_t^{\mathrm{rec}}, \mathrm{hashes}). \tag{137}$$

For partial observation, extend it through GEMLA-Belief-State:

$$\begin{aligned} \mathrm{Pkt}_t^B = \big(&\mathrm{Pkt}_t, \mathcal{B}_t, M_t, C_{\mathrm{obs}}, \Sigma_I, \\ &U_{\mathrm{obs}}, U_{\mathrm{state}}, U_{\mathrm{model}}, U_{\mathrm{dec}}, T_{\mathrm{safe}}, \mathcal{V}_t\big). \end{aligned} \tag{138}$$

For the PDE program, extend it to

$$\mathrm{Pkt}_t^{\mathrm{PDE}} = (\text{native contract}, P, \Phi_P, \pi_P, J_P^{\mathrm{org}}, \mathcal{N}_P, d_{\mathcal{T}}, \text{boundary margin}). \tag{139}$$

31.1 Logs and proofs serve different roles

Logs support reconstruction, while proof obligations separately establish decision preservation, causal mechanism, posterior calibration, or non-reachability. The implementation must distinguish telemetry, evidence, certificate, and authorization. If all layers run with the same administrator credential and writable configuration, a formal diagram fails to create non-self-escalation or independently reachable recovery.

32 Discussion: What FrontierGEMLA Would Contribute

The integrated architecture turns bounded operation into a staged development program:

$$\begin{aligned}\text{embodiments} &\to \text{belief states} \to \text{shared trajectories} \to \text{objects} \\ &\to \text{minimum transformation geometry} \to \text{DPRI obligations} \\ &\to \text{operators} \to \text{shared skills} \to \text{meta-reasoning} \to G4,\end{aligned} \tag{140}$$

followed by

$$\text{self-observation} \to I_{\text{self}} \to \text{candidate improvements} \to \text{shadow lab} \to \text{audit} \to \text{promotion}. \tag{141}$$

The new PDE and Bayesian telemetry sections sharpen the embodiment interpretation. The PDE program describes how finite observation induces transport over invariant states, while the Bayesian layer lets prediction continue under missing data without laundering uncertainty into certainty.

The contribution is therefore a set of coupled reduction, transfer, inference, and governance tests. Efficient Observation, evidence selection, quantization, Adaptive Compute, PDE-layer invariant transport, Bayesian missing-telemetry inference, and skill compilation each attack a different burden. DPRI organizes those reductions around sufficient obligations of the declared decision. Minimal Transformation Geometry asks how little representational structure is sufficient for decision preservation and transfer. Throughout, CI-1–CI-5 keep capability growth, persistent promotion, audit, recovery, provenance, and authority on separately testable paths.

33 Future Work

We have identified non-exhaustive list stated as explicit research areas with associated proof or experiment obligations.

33.1 Existence and minimality of useful invariants

Efficient Observation proves a conditional principle: if a decision-preserving closed invariant exists, full reconstruction is unnecessary. Its applicability depends on establishing the existence and adequacy of such an invariant for the target reasoning or physical class.

33.2 True weighted-fill-distance guarantees for FrontierGEMLA evidence

DA-FPS proves an approximation guarantee for a specific estimated weighted-fill-distance objective under its assumptions. FrontierGEMLA must determine whether invariant-space density, certification-gap weights, consequence weights, and DPRI obligation terms estimate a meaningful true audit objective.

33.3 Well-posedness of the invariant PDE

Equation (47) requires a theory of existence, stability, and discretization on the invariant manifold. Future work should characterize admissible v_P, J_P^{org}, D_P, $\mathcal{R}_P$, and $\mathcal{G}_P^{\text{miss}}$; determine when q_P has weak solutions; and prove which resolution limits preserve nonvanishing organization.

33.4 Posterior calibration under missing telemetry

The Bayesian missing-telemetry layer requires calibrated likelihoods, masks, priors, and phase-critical observability weights. Future work should compare Kalman, extended/unscented Kalman, particle-filter, Gaussian-process, and neural state-space implementations against real and simulated MoldPredict data.

33.5 Partial identification at ASI scale

The identified state set S_t may itself be too large to represent exactly. FrontierGEMLA needs tractable outer approximations that preserve the conservative direction: approximation must preserve or enlarge the identified set in a way that does not fabricate robust admissibility.

33.6 Existence of cross-domain cognitive invariants

The central mathematical question asks when a common representation of reasoning preserves success, transfer, recovery, and audit distinctions across heterogeneous task families and generalizes beyond one benchmark distribution.

33.7 Minimum transformation geometry and architectural necessity

A formal theory is needed for deciding when a relational/projective-candidate, affine, or metric structure is the minimum sufficient geometry for a declared decision class, and when added architecture is necessary for held-out transfer, robustness, or decision preservation.

33.8 Stability of second-order improvement

Second-order improvement requires conditions under which changes to R_k improve search efficiency without increasing unsafe proposal rates, weakening audit coverage, or overfitting the meta-benchmark.

33.9 Belief-state transfer and domain-independent skill semantics

The four-embodiment program requires tests that separate genuine shared skills from adapters that merely dispatch to unrelated domain routines. Future work should establish conditions under which belief-state operators preserve decision and recovery semantics across physical, financial, governance, and fabrication settings; quantify transfer error under held-out missingness mechanisms; and determine when a compiled skill can be promoted into the common operational library.

34 Conclusion

FrontierGEMLA is proposed here as both a bounded operational runtime and an evidence-bearing route from heterogeneous narrow embodiments toward a conditional S3 bounded-superintelligence candidate. Cross-domain generality must be earned through collision-tested representations, transferable DPRI obligation structure, operator and skill transfer, meta-reasoning, held-out G4 testing, self-invariant discovery, proof-carrying promotion, and second-order process tests.

The operational backbone is

$$\begin{aligned} h_{<t} &\to \text{open cognition} \to Q_\theta \to \text{DPRI} \to \text{bounded knowledge seeking} \\ &\to A_{\text{op}} \to \text{persistent-change boundary} \to \Pi_k^{\text{prom}} \\ &\to \mathcal{A}_{\text{audit}}^{[q]} \to \kappa \to \text{environment}. \end{aligned} \tag{142}$$

with raw-observation provenance and recovery retained as protected side channels. Its computational cascade is

$$N \to r, \qquad n \to b, \qquad r \to q, \qquad B \to B_{\text{eff}}, \qquad H \to H_{\text{eff}}. \tag{143}$$

A complementary embodiment cascade is

$$\begin{aligned}\text{native dynamics} &\to \text{masked observation} \to \text{GEMLA-Belief-State}\\ &\to \text{posterior invariant transport} \to \text{shared skill / Adaptive Compute}\\ &\to \text{admissibility/authority verdict.}\end{aligned} \tag{144}$$

The strongest ASI-facing claim therefore rests on preservation of CI-1–CI-5 together with T1–T10 and R1–R4 as capability, transfer, PDE-layer observation, belief-state inference, shared-skill transfer, and architecture optimization increase.

A Operational Runtime Algorithm Compendium

Algorithm A.1 (Evidence-Geometry Update)**.** *For each evidence region, partition evidence into certified, uncertified, synthetic/imputed, and unresolved classes; compute certified coverage and synthetic opacity; update identified state set and admissibility intervals; emit a signed packet with provenance, coverage, opacity, robust set, and authority mode.*

Algorithm A.2 (Information-Seeking Recovery)**.** *For each currently authorized information action, estimate expected contraction of the identified set, subtract resource and risk costs, and execute only if the value exceeds the frozen threshold. Otherwise return HOLD or human escalation.*

Algorithm A.3 (Decision under Unresolved Identification)**.** *If the robust action set is empty, return HOLD/recovery. Otherwise compute restricted minimax regret over robustly admissible actions and execute the minimizing action only after hard validators, semantic contract, and scoped authority pass.*

Algorithm A.4 (Universal Interaction Adapter)**.** *Register observable inputs, outputs, branch summaries, compute controls, telemetry, and version identity. Reject undeclared telemetry or action channels from the certification path.*

Algorithm A.5 (Efficient Observation Streaming Update)**.** *Compute candidate* $\widetilde{I}_{t+1} = \Gamma(I_t, \Delta_t)$*; evaluate gates; accept, freeze, reject, or route according to frozen policy; emit an audit packet.*

Algorithm A.6 (Density-Aware Expensive-Evidence Sampler)**.** *Seed selected evidence anchors; greedily add the candidate maximizing density-weighted invariant distance; record provenance and obtain certification through the protected path.*

Algorithm A.7 (Calibrated Quantization and Collision Check)**.** *Assign candidate quotient class by decision, hard-gate, margin, and transport signatures. Search controls for collision pairs. Refine partition if a collision exists; otherwise emit a canonical token.*

Algorithm A.8 (Adaptive Compute Controller)**.** *Construct admissible compute set; estimate decision value minus resource and risk costs; continue only when expected value remains positive and authorized.*

Algorithm A.9 (Two-Calibration Agency Update)**.** *Propose a calibration update; compute geometry and decision-boundary deltas; submit the evidence packet to the frozen audit calibration; activate only on PASS.*

Algorithm A.10 (Proof-Carrying Recursive Self-Improvement)**.** *Generate candidates in the shadow lab; evaluate hidden, adversarial, transfer, semantic, recovery, authority, and resource tests; construct a promotion packet; promote only when all hard gates pass and external activation is authorized.*

Algorithm A.11 (Semantic Convention Transition)**.** *Classify a convention change; run semantic regressions; require proof/counterexample packet for structural changes; forbid self-authorization for normative changes; publish a new registry only through external epoch approval.*

Algorithm A.12 (Scoped Authority and Execution)**.** *Require operational admission, semantic translation, and recovery state; issue signed scoped token only from the protected broker; execute within scope and revoke on completion or anomaly.*

Algorithm A.13 (Protected Recovery Kernel)**.** *Revoke active credentials outside certified recovery; restore last certified calibration/architecture or enter read-only/reduced-compute state; require recertification before privilege restoration.*

Algorithm A.14 (Collective-Agent Closure Check)**.** *Construct cross-agent dependency graph, identify reflective cycles, represent each candidate at the operational boundary, treat consensus as evidence rather than authority, and reject unstable fixed-point or semantic interpretation loops.*

B Notation and Status Table

Symbol	Meaning
$h_{<t}$	interaction history before time t
Ω	ambient observation/state space
$\mathcal{M}_I$	decision-preserving invariant space
P_I	Efficient Observation projection $\Omega \to \mathcal{M}_I$
r	invariant dimension/effective complexity
Q_θ	calibrated decision-preserving quantizer
T_θ	quantized carrier/quotient state space
τ	admissibility threshold or remaining time depending on context
γ	positive decision/audit margin
A_{op}	operational agency algebra
$\mathcal{A}_{\text{audit}}^{[q]}$	frozen audit meta-algebra during epoch q
Π_k^{prom}	proof-carrying promotion packet
κ	scoped authority token
E^C, E^U, E^S, E^X	certified, uncertified, synthetic/imputed, and unresolved evidence
S_t	identified state set consistent with evidence and admitted assumptions
B_{eff}	effective decision-relevant branching factor
H_{eff}	effective planning horizon after skill compilation
$\pi_P(z,t)$	predictive/posterior density over latent or invariant states at resolution P
J_P^{org}	organized invariant flux term
$\mathcal{B}_t^{(d)}$	GEMLA-Belief-State packet for embodiment d
$U_{\text{obs}}, U_{\text{state}}, U_{\text{model}}, U_{\text{dec}}$	observation, latent-state, model, and decision uncertainty
T_{safe}	calibrated horizon for belief propagation before fresh evidence is required
$M_c(\tau)$	MoldPredict telemetry missingness mask

$Z_c(\tau)$	latent machine/process state
C_{obs}	phase-critical observability confidence
Σ_I	posterior invariant uncertainty

C External-Source-to-Claim Map

The Predilytics Efficient Observation research release supplies the measure-theoretic decision-preserving observation foundation used by the present stack, and the GEMLA SDK v1.0.0 release provides the public software lineage for GEMLA experimentation. Density-aware farthest point sampling supplies a published geometric evidence-selection baseline. Universal Artificial Intelligence supplies history-based agency, knowledge seeking, and self-modification stressors. Missing-data and partial-identification literatures supply the conservative distinction between imputation and identification. Kalman filtering, particle filtering, and sequential Monte Carlo supply recursive Bayesian inference machinery for telemetry gaps. Physics-informed neural networks and PDE-constrained learning supply external context for PDE-governed inverse problems, while FrontierGEMLA places its PDE at the invariant-observation layer.

D Disclosure Statement

During the preparation of this manuscript, the authors used generative AI and AI-assisted technologies, including AI agents and deep-research tools, to assist with language editing, structural refinement, mathematical organization, source checking, and LaTeX/document preparation. All AI-assisted output was reviewed, verified, and edited by the authors, who take full responsibility for the architecture, content, accuracy, citations, analysis, validation program, and conclusions.

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